

Superstore Sales Performance Analysis

US Retail Orders, Jan 2015 – Dec 2018 | Data Analyst Portfolio Case Study | Excel-based Analysis

TOTAL SALES	TOTAL PROFIT	PROFIT MARGIN	DISTINCT ORDERS	2018 vs 2017 GROWTH
\$2.30M	\$286K	12.5%	5,009	+20.4%

Business Question

Sales grew 20.4% year-over-year in 2018, but profit margin remains a modest 12.5% overall. Which products, categories, and customers are actually driving profit — and where is discounting quietly destroying it?

Data & Method

9,994 order lines / 5,009 orders / 793 customers (2015–2018) cleaned and modeled in Excel with live formulas: SUMIFS/COUNTIFS aggregation, FORECAST() time-series projection, and RFM (Recency-Frequency-Monetary) customer scoring using QUARTILE-based segmentation.

6-Month Forecast

Linear trend projection points to sales climbing from ~\$70.0K (Jan 2019) to ~\$74.5K (Jun 2019) per month, continuing the current growth trajectory — a simple trend line, not adjusted for seasonality or promotions.

Key Findings

- **Furniture is a margin trap:** Furniture sales (\$742K) nearly match Technology (\$836K), but its profit margin is just 2.5% vs. Technology's 17.4% and Office Supplies' 17.0% — discounting is eating almost all of Furniture's profit.
- **A handful of products cause outsized losses:** Three 3D printers and a laser printer alone lose over \$17K combined (e.g. Cubify CubeX 3D Printer Double Head: −\$8.9K profit on \$11.1K sales) — these are priced or discounted below viability.
- **West and East regions carry the business:** West (\$108K profit) and East (\$92K) generate 76% of total profit, while Central lags at just \$40K despite \$501K in sales — a similar discount/margin issue as Furniture.
- **Half the customer base needs attention:** Of 793 customers, only 39 are "Champions" while 396 (50%) fall into "Needs Attention" or "At Risk / Lost" — a significant retention gap versus acquisition-only growth.

Recommendations

- **Cap or restructure Furniture discounts** above a defined threshold (e.g. 20%); re-price the identified loss-making SKUs (3D printers, specific laser printers) or discontinue if margin can't be fixed.
- **Investigate Central region pricing/discount policy** to close the gap with West/East, which generate comparable or lower sales but far higher profit.
- **Launch a win-back campaign for the 396 "Needs Attention" / "At Risk" customers** (targeted offers, check-in outreach) before acquisition spend, since retaining existing customers is typically cheaper than acquiring new ones.

Full analysis: *Superstore_Sales_Analysis.xlsx* (Executive Dashboard, Monthly Trend & Forecast, Product Performance, Customer RFM Segmentation) | Source data: *Superstore_Orders_Clean.csv*